

Unit 2: Earth's Place within the Cosmos

Reading: pages 22–38 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. When astronomers observe redshift in the light coming from a distant galaxy, they know that
 - a) the galaxy is moving away from Earth
 - b) the galaxy is heating up
 - c) the galaxy contains a black hole
 - d) the galaxy is made mostly of hydrogen
2. Hubble's Law states that
 - a) the universe will eventually stop expanding and contract
 - b) light always travels at a constant speed
 - c) all galaxies are roughly the same distance apart
 - d) the farther away a galaxy is, the faster it is moving away from us
3. According to the Big Bang theory, the universe began about
 - a) 4.6 billion years ago
 - b) 13.7 billion years ago
 - c) 100 billion years ago
 - d) 380,000 years ago
4. What is the energy source for all stars?
 - a) chemical combustion
 - b) friction caused by gravity
 - c) nuclear fusion
 - d) radioactive decay
5. A group of tens to hundreds of thousands of mostly reddish stars, held tightly together by gravity in a definite spherical shape, is a
 - a) globular cluster
 - b) binary star system
 - c) irregular galaxy
 - d) open cluster

6. Earth sits about how far from the center of the Milky Way Galaxy?
- a) 100 light-years
 - b) 26,000 light-years
 - c) 100,000 light-years
 - d) 2 million light-years

Part B · True or False

2 questions × 1 point = 2 points

7. Elliptical galaxies appear reddish to yellowish because they contain mostly old stars.
TRUE / FALSE
8. The Milky Way is classified as an irregular galaxy.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Astronomers use two different terms: *dark matter* and *dark energy*. Briefly explain what each one is used to account for.
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10. Describe what happens to a typical star like our Sun once it has burned up all of the helium in its core.

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